DAY 1 · SESSION 2 · 13:30 TO 16:00

The Case of Pump A-001

Know the data. Investigate with Copilot.
Check every answer.



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THIS AFTERNOON

Four acts, two labs, one bridge to tomorrow

13:00	**Know the data**	Tables, documents, the security layer, your files
13:45	**The investigation, live**	Acts 1 and 2: find the asset, read the machine, follow the attacker
14:30	**Lab 1: role exercises**	E01 to E16, by role, weak then structured
15:15	**Break**	
15:30	**Trust and tell**	Act 3 and 4: the version trap, three audiences, the team mini project
16:30	**From Copilot to OpenAI**	What changes tomorrow, guardrails, Day 2 preparation

THE STORY

Monday morning at the plant

> *"Is anything at risk this week, and what should we do about it?"*
>
> The plant director, 08:15

128

assets in the plant

8,760

hourly sensor readings for A-001

38,677

log events in 30 days

19

documents, three versions of some

You have Copilot and one afternoon. Let us answer the director, and prove every number.

PART 1

Know the data

ENTERPRISE DATA

Structured and unstructured: most real questions need both

T Structured: tables

Rows and columns with a fixed meaning.

Asset 360, work orders, sensor readings, logs.

Best for counts, filters and trends.

Copilot in Excel works here.

D Unstructured: documents

Policies, procedures, reports, runbooks.

19 documents in your pack.

Best for rules, reasons and procedures.

Copilot in Word and chat today. RAG on Day 2.

A number without the procedure that governs it is not a decision.

One key joins almost everything: asset_id

Know the grain of each table: one asset, one hour, one event, one document.



YOUR PACK

What is in your files, and what each one answers

Workbook	What is inside	Answers questions like
ENEC_Asset_360_Security	Assets with security: priority, vulnerabilities, incidents, alerts, access	Which assets need attention first?
asset_360	Assets, A-001 timeline, documents, hourly sensors	What is the machine telling us?
ENEC_Security_Logs	Logins, network flows, controller commands	Who tried to get in, and when?
ENEC_Security_Log_Windows	15-minute windows with 16 features	What does an attack look like? (Day 3)
project_360	Projects, KPIs, risks, actions	Which risks matter most?
2_Documents_Word	9 A-001 documents and 10 security documents	What does the approved procedure say?

Every sheet is already an Excel table. Copilot works on one workbook at a time.

MEET THE PATIENT

Pump A-001: cooling water, Train A, Critical

97.8

priority score, rank 1 of 128

68.05

health score, risk MEDIUM

3

open security issues, max CVSS 8.2

6

open work orders, 3 high priority

Operations says

Patch overdue, 23 sensor anomalies in 90 days, attention flag Yes.

Security says

Restricted network zone, access risk flag Yes, a security incident in the last 90 days.

All synthetic. One pump carries every exercise for the next four days.

EXERCISE · 10 MINUTES

Five questions to ask any new table

1 What is in it?

rows, columns, grain, date range

2 Is it clean?

missing values, duplicates, odd values

3 What patterns show?

groups, trends, correlations

4 What stands out?

outliers, one asset that differs

5 What is missing?

what the data cannot tell us

STARTER PROMPT · TRY IT ON ANY SHEET

```
Describe this table: rows, columns, grain and date range. List columns with missing values. Then suggest three questions worth asking, and say which columns each one uses.
```

Pick one sheet from your role. Run the starter prompt. Share one surprise.

THE SECURITY LAYER

Same plant, same assets, four kinds of security data

1 Posture

8 new columns on Asset 360
174 vulnerabilities
Access review per asset
Combined priority list

2 Raw logs

14,137 login events
21,492 firewall flows
3,048 controller commands
30 days of activity

3 Detections

478 SIEM alerts from 6 rules
43 incidents
False positives included
Noise to tune

4 Model data

23,452 windows of 15 minutes
16 features from the logs
Attack labels
Used on Day 3

One join key, asset_id. Operations and security finally look at the same list.

PART 2 · LIVE

The investigation

Act 1

Find the problem

Act 2

Investigate A-001

Act 3

Can we trust it?

Act 4

Tell people and scale it

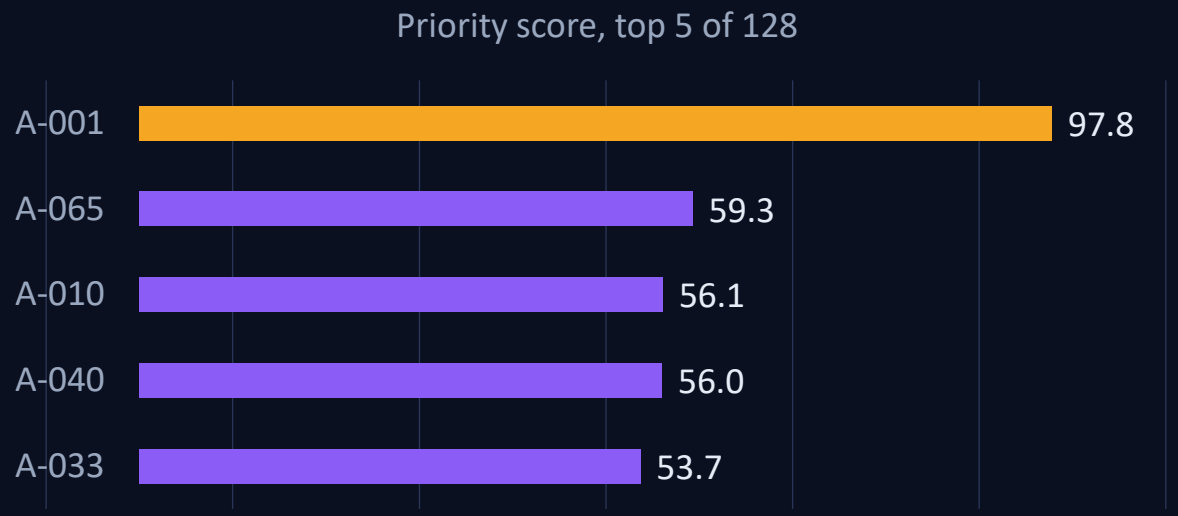
ACT 1 · PROMPT 1 · COPILOT IN EXCEL

Which assets need attention this week?

ENEC_Asset_360_Security.xlsx · sheet Asset_Priority_View

PROMPT

```
You are the plant's reliability and cyber risk
lead. Using this table, list the top 5 assets
by Priority_Score. For each, give asset_id,
health_score, Open_Security_Issues and
Patch_Status, then explain in one sentence why
number 1 ranks first.
```



WHAT SHOULD COME BACK

A-001 first at 97.8, far ahead of A-065 at 59.3. Critical, health score 68.05, 3 open security issues, patch overdue.

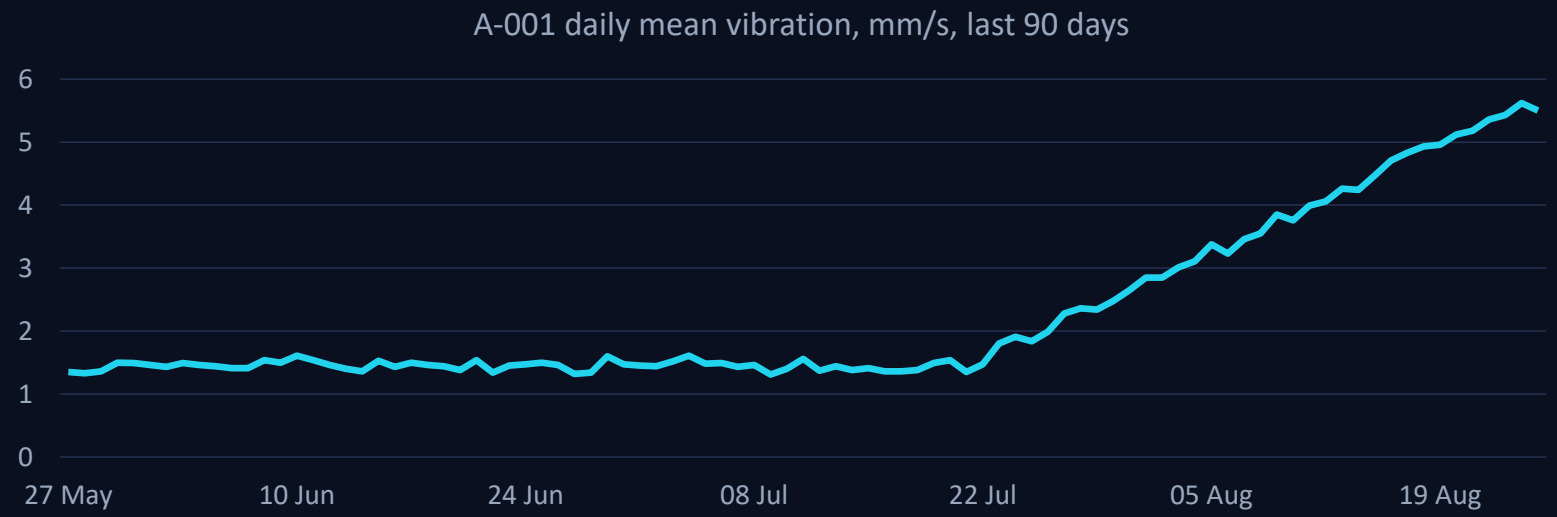
Teaching point: a role and a named output turn a vague request into a table you can check.

What is the machine telling us?

asset_360.xlsx · sheet a001_sensor_hourly · 8,760 rows, allow 30 seconds

PROMPT

```
Plot the daily average vibration for A-001 as a line chart. Tell me the date the rise starts and the first date anomaly minutes appear, and add both as notes beside the chart.
```



1.4 mm/s for weeks

Rise starts ~22 Jul

Anomalies from 18 Aug

ACT 2 · PROMPT 3 · COPILOT IN EXCEL

Who tried to get in?

ENEC_Security_Logs.xlsx · sheet Auth_Logs · about 14,000 rows

PROMPT

```
Find the source address with the most LOGIN_FAILURE events against A-001. How many failures, on what date, and over how many minutes? Describe it in two sentences as a SOC analyst would.
```

203.0.113.45

source address

34

failed logins

~20 min

one short burst

14 Aug

2026

Narrow questions work best on big tables: one asset, one event type.

ACT 2 · PROMPT 4 · COPILOT IN EXCEL

One asset, the whole story

ENEC_Asset_360_Security.xlsx · sheet Security_Incidents

PROMPT

```
Build a timeline of every security incident on A-001: date, type, severity, status. Then say what a human should decide next. Do not recommend actions the data cannot support.
```



Correlation is not cause. The data does not show the login burst caused the vibration.

ONE DATASET, FOUR PERSPECTIVES

The same tables answer different questions for different roles

Operations

Which assets show abnormal readings?

Cybersecurity

Which assets have open issues or exposure?

Projects

Which risks touch critical assets?

Management

What needs a decision this week?

Cross-discipline question: high-criticality assets with abnormal readings and unresolved security issues

5 assets: A-001, A-040, A-033, A-109, A-115 (exercise E03)

Operations alone flags 2 assets. Operations plus security gives a better list.

LAB 1 · 45 MINUTES

Your role exercises: E01 to E16

If you work in	Start with
Cybersecurity, OT assurance	**E04, E05, E08, E10**
Incident preparedness, Technical Security Services	**E07, E09, E16**
Architecture, Integration, Development	**E11, E12, E13**
Operations, Applications	**E03, E14**
Management, Project, Assurance, Delivery	**E02, E06, E15**

For each exercise

1. Weak prompt first.
2. Structured prompt second.
3. Compare with check_against in ENEC_Copilot_Exercises.xlsx.
4. Write: right, fixed, could not know.

Everyone also does E01 and E16. Finished early? Try another role's exercises.

PART 3

Can we trust it? Then tell people.

The version trap

Attach the three patch standards: SEC-PROC-PATCH-001 V1, V2 and V3D

V1

Superseded

Old rules. Still on the drive.

V2

Approved

The only authoritative version.

V3D

Draft

Proposed changes. Not in force.

Watch the first answer

Did it quote V1 or the draft V3D?

PROMPT A

```
What is the deadline for applying critical patches? Answer from these documents and name the document and version you used.
```

PROMPT B

```
Which of these documents is approved? Answer again using only that one.
```

WHAT SHOULD COME BACK

The correct source is V2, the only approved version.

AI reads whatever you give it. Governance decides what it should read.

ACT 3 · PROMPT 6 · ANY COPILOT CHAT

Make Copilot check its own answer

PROMPT

```
List three ways your previous answer could be wrong, and how I could check each one in this workbook in under a minute.
```

1 Wrong filter

Filter the column yourself and count the rows.

2 Wrong grain

Hourly or daily? One asset or all of them?

3 Wrong source

Which sheet or document version did it use?

A habit for life: after every important answer, ask how it could be wrong.

Tell the right people, in their language

PROMPT 7 · ONE FACT, THREE AUDIENCES

```
Rewrite this finding three ways: (1) three lines
for the plant director, (2) a technical note for
the SOC team, (3) a task for the maintenance
planner with what to check first.
```

PROMPT 8 · THE STATUS EMAIL (Security_Incidents)

```
Write a three-sentence status update on the open
incidents, ordered by severity, ending with the
decisions a human must make. Use a formal tone for
senior management.
```

Check

Same numbers in all three versions? If a number changed, catch it.

Check

Mentions SEC-INC-0040 on A-001. Ends with decisions, not actions Copilot claims it took.

A draft in seconds. You sign it, so you check it.

TEAM MINI PROJECT · 30 MINUTES

The A-001 decision brief

Should A-001 be taken offline for inspection this week? Answer on one page, using only Copilot and your pack.

One finding

in one sentence

Three numbers

each with its sheet or document

The main risk

and what is still unknown

A recommendation

citing the approved procedure (V2)

The human decision

who decides, and what they need

Scoring

Correct numbers (priority 97.8, 3 open issues, rise from ~22 Jul, anomalies from 18 Aug, 34 failures from 203.0.113.45). Every claim sourced. The right document version. Three-minute presentation per team.

Write the rules once: your own assistant

PROMPT

```
Write instructions for an assistant that answers questions about pump A-001. It must use only the approved documents, treat V2 as authoritative, cite the document for every answer, and say "I don't know" when the documents don't cover it. Keep it under 150 words.
```



If your organisation allows it, paste these instructions into the Copilot agent builder with the approved documents. If not, keep them: on Day 3 we use them as the agent's system prompt in code.

PART 4

From Copilot to OpenAI

Today you use a model. Tomorrow you build with one.

1 Today: Microsoft Copilot

An assistant built on OpenAI models.
Works inside Excel, Word and chat.
You write prompts; Microsoft decides the rest.
Great for analysis and drafting.

2 Days 2 to 4: the OpenAI API

You call the model from Python in Google Colab.
You choose the documents, the rules and the tools.
Models: gpt-5.4-mini, with gpt-5-mini and gpt-4o-mini as fallbacks.
Embeddings: text-embedding-3-small. Framework: LlamaIndex.

Same model family. The difference is control: tomorrow you decide what it reads and what it may do.

THE BRIDGE

Your Day 1 prompt becomes the system message on Day 2

TODAY · IN COPILOT

```
Role: You are an OT security analyst.
Evidence: Use only the approved documents.
Check: Cite the document. Say "not in the
data" if unsure.
```



TOMORROW · IN COLAB

```
from openai import OpenAI
client = OpenAI()   # key from Colab Secrets

resp = client.chat.completions.create(
  model="gpt-5.4-mini",
  messages=[
    {"role": "system", "content":
SEVEN_PART_PROMPT},
    {"role": "user", "content": question}])
```

What stays the same

The seven parts, the checks, the human decision.

Everything you practised today is reused in code. Nothing is wasted.

GUARDRAILS

Rules for everyone, every day, from today

1 Synthetic data only

No real ENEC data, names, systems or incidents in any AI tool.

2 Check before you trust

Compare with a filter. Cite the document and version.

3 A person decides

No exercise acts on a control system. Agents read and recommend only.

4 Keys stay secret

Your OpenAI key goes in Colab Secrets, never in a cell or on GitHub.

5 Official accounts

Copilot with your Microsoft account. Colab with a Google account.

6 Say what you do not know

"Unknown" is a good answer.

LAB 2 · 15 MINUTES

Find one practical AI use case for your role

Start with the decision, not the technology.

1 Decision

what is decided, by whom

2 Evidence

which data and documents

3 Risk

what if it is wrong

4 Value

time, quality, risk reduced

5 Human step

who reviews

Hand in

One use case you could defend to your manager, on one card. We keep them and come back to them on Day 4.

Time saved is only one measure. Ask what it costs when the decision goes wrong.

DAY 1 TO DAY 2

What today gave you, and what to do tonight

What Day 1 gave you

Fluent use of Copilot, with checks.
A seven-part prompt you can reuse.
A map of the data and the documents.
A decision brief on A-001.
One use case for your role.

Tonight, 10 minutes

Sign in to Google Colab with a Google account.
Have your OpenAI key ready for Colab Secrets.
Check you can open the course repository on GitHub.
Open Day2_00 and run the first cell.

Group photo before you leave. Tomorrow: build a RAG application on trusted A-001 documents.